L2.2 Cramer's Rule

Transcript Language

English

Hello and welcome to the maths 2 component of the online BSC degree on Data Science. In today's

video we are going to look at Cramer's Rule which employs the determinant in order to find

the solutions of a system of linear equations, when the coefficient matrix is invertible.

Let us go on and do an example.

So, let us look at this linear system, so 4x1 minus 3x2 is 11 and 6x1 plus 5x2 is 7.

So, let us recall what was the matrix representation. So, that is Ax is b, so

the matrix A is given by the coefficients, so 4, minus 3, 6, 5 and b is the constants on the right,

so it is a column vector or column matrix 11, 7 and x is the unknowns x1, x2.

So, this it is easy to compute this solution, it has a unique

solution x1 is 2 and x2 is minus 1. Let us quickly go through how, so we can multiply

the first one by 3 to get 12x1 minus 9x2 is 33,

we can multiply the second one by 2 to get 12x1 plus 10x2 is 14 and then we subtract

let us say the first one from the second one, so in that case we get 19x2 is

so what should we get, so 14 minus 33, so that is minus 19, that tells us x2 is minus 1 and then

we can substitute in one of these expressions and get that x1 is 2, one of these equations

and get that x1 is 2. So, we can easily check that this is a unique solution.

So, now let us do something slightly different and reinterpret this using determinants.

So, here is the same example and here is how we will apply Cramer s rule.

So, we look at the coefficient matrix A, so that was 6, minus 3, sorry, 4, minus 3,

6 and 5. Let us compute the determinant of A, so if we do that that is 4 times 5

minus negative 3 times 6 so that is 20 plus 18 which is 38, let us keep that value in

mind. So, the determinant of A is 38. So, replace the first column of A by the

column vector b and we will call that matrix A subscript x1 because the first column are

the coefficients corresponding to x1 in the two equations. So, A subscript x1 is

11, minus 3, 7, 5. So, we have replaced the 4, 6 over here by 11, 7. Replace the second column of

A by the column vector b and call it x2, Ax2, so A subscript x2, so we are doing the same thing

for the second column. So, now the first

column is the same as the first column of A and the second column is replaced by 11, 7 so we get

4, 11, 6, 7 let us say subscript x2, let us calculate determinant of Ax1, so you can do this,

this is 11 times 5 minus negative 3 times 7 so that works out to 76

and then let us calculate the determinant of x2, sorry, Ax2, A subscript x2. So, if you do

that that is 4 times 7 minus 11 times 6 that is minus 38, so remember that the determinant of

A was calculated to be 38. So, let us keep this in mind 38, 76 and minus 38.

So, now what do we do? So Cramer's rule will say, let us

calculate determinant of A subscript x1 by determinant of A and determinant of A subscript x2

by determinant of A. So, let us do that, so this is 76 by 38 and this is minus 38 by 38

and these are exactly the solutions. So, x1 is

2, remember that is what we got in the first slide and x2 is

minus 1 which is exactly what this is. So, at least in this example we have

this strange method using determinants and it gives us the solutions that we require,

that we should have got, not required. So, let us study what is Cramer's rule in general, so this is for when the coefficient matrix is an invertible matrix,

so what do you mean by invertible? Invertible means that the inverse exists.

So, consider the following system of linear equations of 2 variables a11x1 plus a12x2 is b1,

a21x1 plus a22x2 is b2. So, the matrix representation here is a11, a12, a21, a22

and b is b1 the column vector b, b1, b2. And the assumption here is that the matrix

A is invertible, meaning its inverse exists and because its inverse exists, remember that in the

previous video with determinants we have seen that determinant of A inverse is 1 by determinant of A

which in particular means that determinant of A is non-zero, so 1 by determinant of A exists.

So, let us define these 2 matrices as in the previous example. So, we replace the first column

by the vector b1, b2 and we call that new matrix A subscript x1 and we replace the second column

by the column vector b and we call that new matrix A subscript x2.

And then Cramer's rule says the solution of the system of equations and I am saying the which means it is a unique solution, in two variables, so this is determinant of A

subscript x1 by determinant of A, note that this division makes sense because its

invertible and x2 is determinant of A subscript x2 by determinant of A, again makes sense and we know

explicitly what these expressions are, for 2 by 2 matrices the determinant is easy to compute.

So this is the numerator on for the first expression is b1 a22 minus a12 b2

and the numerator for the second expression is a11 b2 minus b1 a21

and the determinant of the matrix A is a11 a22 minus a21 a22.

So, that is the algorithm you have to follow for Cramer's rule to find the

solutions of a system of equations where the coefficient matrix A has non-zero determinant.

We can make this more general, so let us do a 3 by 3, see

Cramer's rule for 3 by 3 matrices. So, again here the coefficient matrix must be invertible,

so the determinant of A must be non-zero because we divide by the determinant. So, here is the system of linear equations in three variables, so recall that its matrix representation is Ax is

b, where A is the coefficient matrix and b is this column vector of constants b1, b2, b3.

So, in that case what is Cramer's rule? So, we will have to define these new

matrices, again 3 by 3 matrices, so A subscript x1 is where you replace the first column of

A by the column vector b, A subscript x2 is where you replace the second column vector,

sorry, the second column of A by b and A subscript x3 is where you replace the third column vector,

third column of A by b. And then how do we get the solution?

So, the solution meaning again its unique of the system of equations that we saw is x1 is

determinant of A subscript x1 by determinant of A, x2 is determinant of A subscript x2 by determinant

of A and x3 is determinant of A subscript x3 by determinant of A. Again very important is that

the determinant here must be non-zero, otherwise of course, we cannot divide and assuming that A

has an inverse ensures this or explicitly we can say that determinant of A is non-zero.

So, how did we define these A subscript x1, A subscript x2, A subscript x3?

So, in the first one we the first column was replaced, in the second one the second

column was replaced and in the third one the third column was replaced. So, these

and the notations should tell you which columns are replaced because x1 means the coefficients

corresponding to x1 that is the column which is replaced, so those were a11, a21 and a31.

Similarly, for x2 and x3. So, this is Cramer's rule for 3 by 3 matrices or for a system of

linear equations in three variables, maybe let us do a quick example.

So, consider this system of equations where the coefficient matrix, so I am instead of

writing down the system I am writing down the coefficient matrix and the constants.

So, the coefficient matrix is 1, 0, 3, 0, 2, 5, 4, 3, 1 and the constants are 0, 2 and 1 respectively

for the three equations. So, to use Cramer s rule first let us look at the determinant.

So, the determinant of A, so we will have to compute this, so the determinant is minus 37,

so I encourage you to compute this. So, since it is non-zero we can apply Cramer s rule.

So, let us follow the next steps in the procedure for Cramer's rule.

So, we have to compute these matrices A subscript x1, A subscript x2 and A subscript x3.

So, how do we compute these matrices? We replace the corresponding columns by the

column b. so, here we get 0, 2, 1 so b, this is b and the other things remain same here we get

0, 2, 1 in the second, in place of the second column and here we get 0, 2,

1 in place of the third column. So, these are our matrices

A subscript x1, A subscript x2 and A subscript x3 and then we compute their determinants, so the

determinant of A subscript x1 is 12, you can check this. So, here we have first row has two 0s, so

this is an easy computation similarly, for A subscript x2 you can check this is minus 27 and A

subscript x3 the determinant is minus 4. And then by Cramer's rule

we get that the solution to this system of equations is x1 is minus 12 by 37, x2 is 27 by 37 and x3 is 4 by 37. So, I will encourage you to

substitute these x1, x2, x3 and check that indeed this is the solution and you can,

you know do a usual method of solving equations and check that indeed this is the solution.

So, finally let us end with Cramer's rule for the n by n, so now if you have

n systems in, a system of n linear equations in n unknowns. So, remember here that we should have,

the coefficient matrix has to be square first of all, so we need n equations in n unknowns,

this will not work otherwise and the second thing we need here is that

the coefficient matrix is invertible or that the determinant is non-zero.

So, this is the general system of equations, A is this matrix of coefficients aij, a11, a12,

a etcetera and then b is the constant, so column vector with b1, b2, bn, then you define Axi that

is A subscript xi, this is a new n by n matrix and how do we obtain this? We obtain this by for the i-th column you substitute the, you replace it by the column vector b. So, if you do that

then you compute its determinant and then the solution to the system of linear equations.

So, Cramer's rule states that it is given by xi is determinant of A subscript xi by determinant of A,

so we compute for each i. So, x1 is determinant of A subscript x1 by determinant of A, x2 is

determinant of A subscript x2 by determinant of A and so on up till xn is determinant of

A subscript xn by determinant of A, that is what Cramer's rule states.

So, in Cramer's rule remember that we need, this is very important n equations in n unknowns,

that is when we can apply it and further we need that the determinant of the coefficient matrix

A is non-zero and then we can explicitly work out what the solution is. Thank you.